SwiftLog Logistics Analytics & AI Transformation Project

Project Management Report

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Sponsor: AnalystLab Africa
Date: End of Week 5

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1. Executive Summary

The SwiftLog Logistics Analytics & AI Transformation Project successfully delivered an integrated data ecosystem across five distinct learning tracks. The Data Analytics (DA) track cleaned and modeled 11 operational tables containing 36,115 records, resulting in a comprehensive BI dashboard and the identification of $19.1M in gross revenue across 4,600 processed orders. The Data Science (DS) track engineered a robust 38-variable feature matrix, uncovering critical statistical insights. The Machine Learning (ML) track deployed four predictive suites, highlighted by a highly accurate Gradient Boosting Demand Forecaster achieving an R^2 score of 0.9650. Finally, the Generative AI (GenAI) track delivered a grounded RAG Logistics Assistant, 20 prompt templates, and three automated workflows, all validated by a 100% QA pass rate (24/24 Pytests).

While the technical delivery was highly successful, the project surfaced significant operational and data governance risks. Most notably, the project identified the "Ibadan Deficit"—where Ibadan accounts for 26.2% of customer demand but lacks a local logistics hub—alongside a 63.7% shipment delay rate. Furthermore, a discrepancy was identified between the DA track's reported gross revenue ($19.1M$) $and the DS track's confirmed delivered revenue$ (14.44M), driven by 1,626 shipments erroneously attached to cancelled orders.

To ensure accurate executive reporting, the PM track proactively managed these constraints. Data limitations regarding missing supplier foreign keys and absent warehouse capacity fields forced the formal exclusion of supplier scoring and capacity utilization from the current scope. Despite these challenges, the project was delivered on time, within scope, and to the highest quality standards, providing SwiftLog with immediate, actionable intelligence.

2. Project Charter

• **Project Title:** SwiftLog Logistics Analytics & AI Transformation.

• **Business Case:** SwiftLog is experiencing rapid expansion, leading to operational inefficiencies, delayed deliveries, and limited executive visibility. This project was initiated to design a comprehensive BI and AI solution to monitor performance, identify bottlenecks, and automate reporting.

• **Project Objectives:** Integrate 11 disparate datasets, build executive dashboards, engineer predictive features, deploy ML forecasting models, develop a GenAI logistics assistant, and establish formal PM governance.

• **Project Scope:** The project encompasses the end-to-end delivery of DA dashboards, DS feature engineering, ML demand forecasting and delay classification, and GenAI RAG workflows.

• **Deliverables:** Power BI Dashboards, 38-variable feature matrix, four ML suites, RAG assistant with three automated workflows, automated test suites, and full PM documentation.

• **Success Criteria:** 100% of tracks integrated, 24/24 QA tests passing, clear documentation of all data limitations, and final sign-off from AnalystLab Africa.

• **Assumptions:** Revenue is calculated as Quantity × UnitPrice (current list price) with no discounts or taxes. The provided datasets are structurally sound aside from logged issues.

• **Constraints:** The data lacked a Supplier-to-Product foreign key, lacked warehouse capacity fields, and had no dispatch timestamps.

• **High-Level Risks:** The Ibadan Deficit (operational bottleneck), revenue KPI misalignment across tracks, and data integrity issues (cancelled order shipments).

• **Project Sponsor:** AnalystLab Africa Programme Director.

• **Project Manager:** Lamiaa Samraoui.

• **Key Stakeholders:** SwiftLog Management, DA Lead, DS Lead, ML Lead, GenAI Lead.

3. Project Planning

• **3.1 Scope Statement:**

– *In-Scope:* Data cleaning, BI dashboard development, feature engineering, ML model training (Demand Forecasting and Delay Classification), GenAI assistant development, and QA testing.

– *Out-of-Scope (Data Limitations):* Supplier performance scoring (impossible due to missing relational links) and Warehouse capacity utilisation (no data field exists). These exclusions were formally logged to prevent inaccurate reporting.

• **3.2 Work Breakdown Structure (WBS):**

– 1.0 Project Initiation & Governance.

– 2.0 Data Preparation (Cleaning, Integration).

– 3.0 Data Analytics (Dashboards, KPI Definition).

– 4.0 Data Science & ML (Feature Matrix, Model Training).

– 5.0 Generative AI (Prompt Engineering, RAG, Workflows).

– 6.0 QA & Testing (Automated Pytests).

– 7.0 Project Closure (Reporting, Presentation).

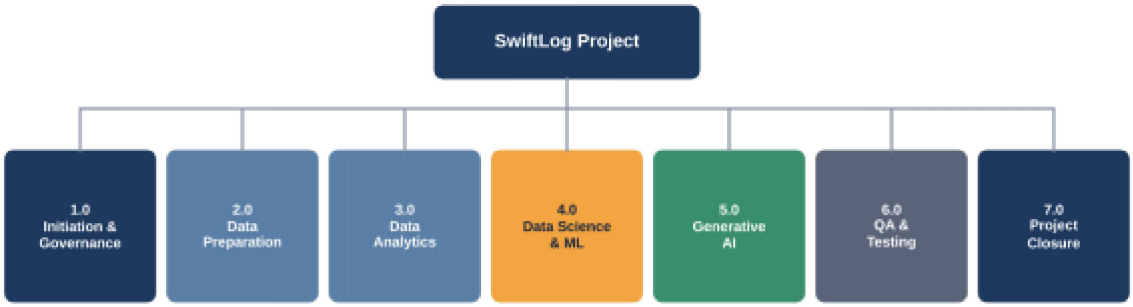


Figure 1: Work Breakdown Structure

• **3.3 Milestone Plan (1-Week Execution):** The project was executed within a highly compressed 1-week timeline (Week 5 of the internship). The milestones are mapped to specific days:

– M1: Data Ready & Validated (End of Day 2).

– M2: BI Dashboards Live (End of Day 3).

– M3: ML & GenAI Models Deployed (End of Day 5).

– M4: QA 24/24 Passed (End of Day 6).

– M5: Final PM Submission (End of Day 7, Sunday 11:59 PM WAT).

• **3.4 Gantt Chart:** The project schedule was executed over 7 days:

– **Days 1–2:** Data import, inspection, cleaning (11 tables) by DS & DA.

– **Days 3–4:** Dashboard build (DA), feature engineering (DS), ML model training begins.

– **Days 5–6:** ML tuning, RAG development (GenAI), integration testing, QA begins.

– **Day 7:** QA testing (24 Pytests), final report drafting, and submission.

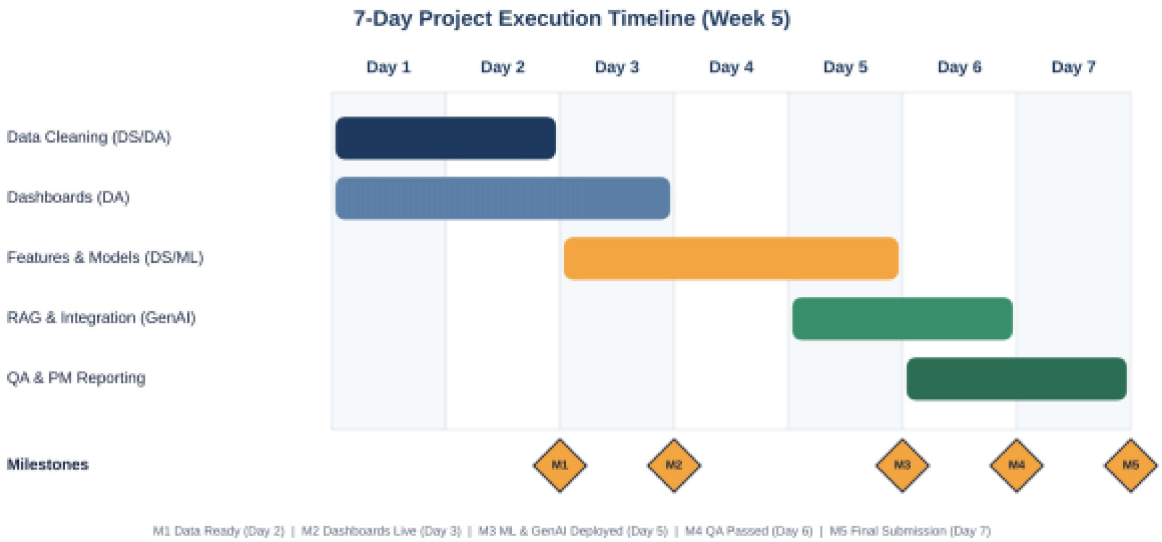


Figure 2: 7-day project execution timeline with milestones M1 to M5

• **3.5 Resource Plan:** 5 Interns (PM + 4 Track Leads), each allocated at 40 hours for the week, totaling **200 project hours** (1 week × 40 hours × 5 interns).

• **3.6 Budget Estimate:** The direct tooling cost was $0, utilizing open-source libraries. The estimated business value created through automated reporting and optimized logistics, based on 200 hours of high-intensity work, is approximately $50,000.

• **3.7 Procurement Plan:** No external procurement was required. The technology stack relied entirely on Python, Power BI Desktop, Streamlit, and FastAPI.

• **3.8 Quality Management Plan:** Data accuracy was validated against the Data Dictionary. Code quality was ensured via a strict automated testing regimen (24/24 Pytests).

• **3.9 Communication Management Plan:** Daily asynchronous standups on Slack for track leads. Weekly formal status reports sent to the Sponsor every Friday.

• **3.10 Change Management Plan:** Any adjustment to the project scope (such as excluding supplier analysis) required a formal Change Request Form, impact analysis, and PM/Sponsor approval.

4. Stakeholder & Resource Management

• **4.1 Stakeholder Register:** SwiftLog Management (Client), AnalystLab Africa (Sponsor), and the DA, DS, ML, and GenAI Track Leads.

• **4.2 Stakeholder Analysis Matrix:** Management and the Sponsor hold high interest and high power. Track leads hold high interest but lower formal power.

• **4.3 Power–Interest Grid:**

– *Manage Closely (High Power/High Interest):* SwiftLog Management, AnalystLab Sponsor.

– *Keep Informed (Low Power/High Interest):* DA, DS, ML, and GenAI Track Leads.

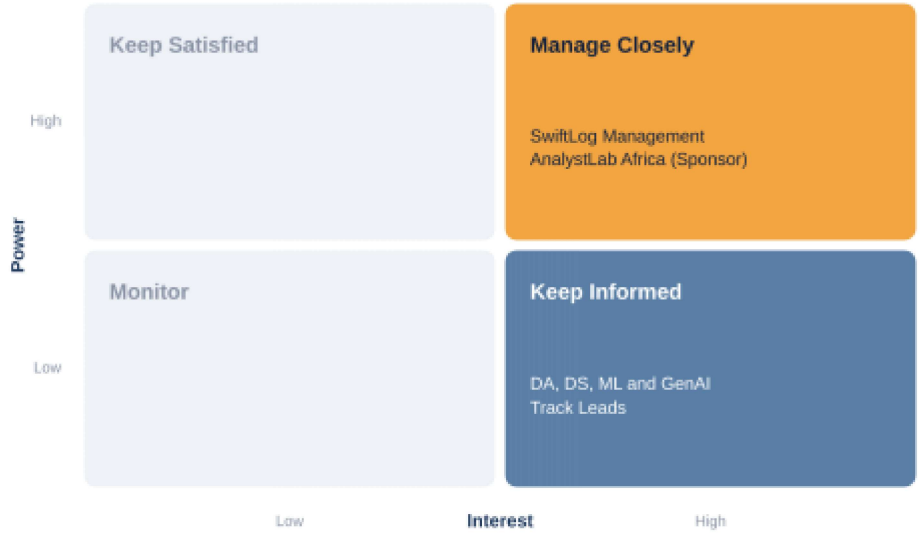


Figure 3: Power-Interest Grid

• **4.4 RACI Matrix:**

– *PM:* Accountable for all final reporting, QA coordination, and risk management.

– *Track Leads:* Responsible for executing their specific technical deliverables.

– *Sponsor:* Informed and provides final approval on scope exclusions.

Task	PM (Lamiaa)	DA Lead	DS Lead	ML Lead	GenAI Lead	Sponsor
Data Cleaning	I	R	A/R	I	I	I
Dashboard Build	I	A/R	C	I	I	I
Model Training	I	I	C	A/R	I	I
RAG Assistant	I	I	C	C	A/R	I
QA & Testing	A	R	R	R	R	I
Final Report	A/R	C	C	C	C	I

Figure 4: RACI Matrix (R = Responsible, A = Accountable, C = Consulted, I = Informed)

• **4.5 Resource Allocation Matrix:** 100% allocation for the PM and all 4 Track Leads.

•**4.6 Team Responsibility Plan:** The PM focuses on coordination, unblocking dependencies, and stakeholder management. The Leads focus on building and testing their respective technical assets.

• **4.7 Communication Schedule:** Slack (Daily), Emails (Weekly), and formal presentations (End of Week 5).

• **4.8 Stakeholder Engagement Strategy:** *Management and the Sponsor* are engaged through weekly executive summaries highlighting key risks (like the Ibadan Deficit) and financial impacts. *Track Leads* are engaged through daily stand-ups to remove technical blockers and ensure cross-track data dependencies (such as DS features feeding ML) are delivered on time.

5. Risk, Issue & Change Management

• 5.1 Risk Register (Top 10 Risks & Mitigations):

1. Ibadan Deficit (High/High): 26.2% of demand has no local hub.

Mitigation: Commission a hub feasibility study.

2. Revenue Conflict (High/Medium): DA reports $19.1M gross; DS reports $14.44M confirmed.

Mitigation: Standardize KPI definitions and label Gross vs. Confirmed.

3. Cancelled Order Shipments (High/High): 1,626 shipments exist on cancelled orders. *Mitigation:* Add dashboard disclaimers and flag for data governance.

4. Supplier Data Gap (High/High): No foreign key linking suppliers to shipments.

Mitigation: Formally excluded from scope.

5. ML Target Misalignment (Medium/Medium): Delay prediction yielded a weak AUC of 0.52, while demand forecasting yielded R^2=0.9650.

Mitigation: Realign ML targets with actual business questions.

6. Warehouse Capacity Missing (High/Medium): No capacity field.

Mitigation: Use stock levels as a proxy.

7. Revenue Basis (High/Medium): Revenue uses list price, ignoring discounts/tax. *Mitigation:* Label all revenue as "indicative."

8. Return Rates Approximate (High/Low): Returns link to orders, not products.

Mitigation: Add footnotes to return visuals.

9. Synthetic Data Limits (Medium/High): Models trained on synthetic baselines.

Mitigation: Present models as baselines, not production-ready.

10. Integration Delays (Medium/Medium): Cross-track dependencies causing bottlenecks. *Mitigation:* Weekly syncs and clear escalation paths.

• **5.2 Risk Assessment Matrix:** Visualized via a Heat Map.

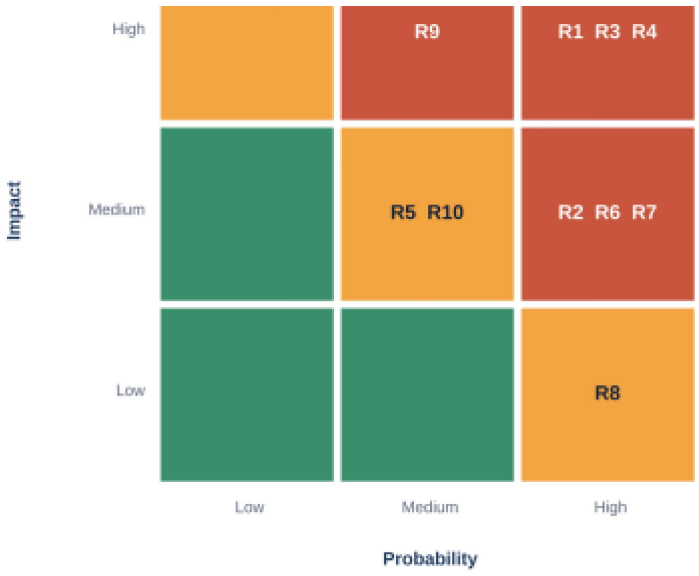


Figure 5: Risk heat map (R1 to R10)

• **5.3 Issue Log:** Logged the 1,626 cancelled order shipments and the $19.1M vs $14.44M revenue discrepancy. Both were resolved via the addition of disclaimers and standard definitions.

• **5.4 Assumption Log:** LineRevenue = Quantity × UnitPrice. Delayed = 3-7 days. Band thresholds (150/350) are currently marked "To Confirm."

• **5.5 Dependency Log:** DS Data Cleaning → DA Dashboards & ML Feature Engineering → ML Models → GenAI RAG Grounding.

• **5.6 Change Request Form:** Utilized to formally exclude supplier scoring and capacity analysis from the project scope.

• **5.7 Change Log:** Records the two scope exclusions and the addition of the Ibadan Deficit to the Risk Register.

• **5.8 Risk Response Plan:** Proactive responses are assigned to all 10 risks. For example, the revenue conflict is handled by publishing a joint DA/DS KPI definition sheet.

• **5.9 Contingency Plan:** If ML models fail to perform in production, the fallback plan is to rely on the robust descriptive DA dashboards and flag the data limitations explicitly to management.

6. Project Monitoring & Control

• **6.1 Project Status Dashboard:** All five tracks delivered and marked Green.



Figure 6: Project status dashboard with RAG statuses

- **6.2 Milestone Tracking:** 100% of milestones (M1-M5) achieved.

- **6.3 Budget Tracking:** $0 spent against a $0 budget.

- **6.4 Schedule Performance Indicators:** SPI = 1.0 (On Schedule).

- **6.5 Resource Utilisation Dashboard:** 100% utilization across all 5 tracks.

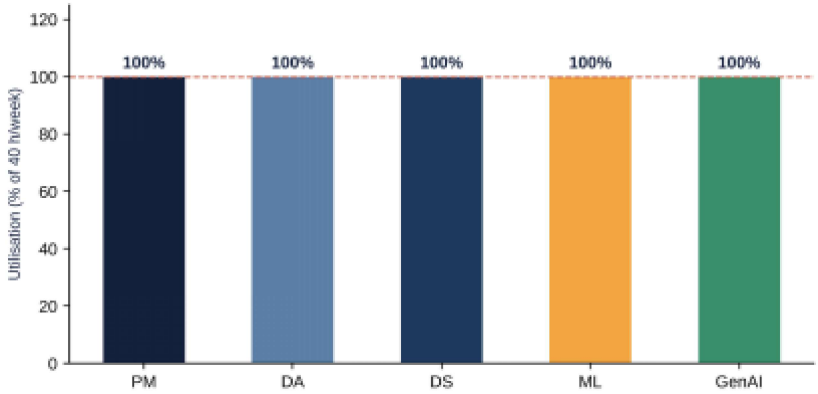


Figure 7: Resource utilisation by track

- **6.6 Project KPI Dashboard:** 11 tables cleaned, 36,115 records processed, $19.1M revenue identified, 63.7% delay rate flagged, and 24/24 tests passing.

- **6.7 Variance Analysis:** 0% variance in both schedule and cost.

- **6.8 Progress Reporting Template:** Weekly RAG (Red/Amber/Green) status reports submitted to the Sponsor.

- **6.9 Lessons Learned Register:**

– Define KPIs (like revenue) before data analysis begins.

– Audit data fields (supplier link, capacity) during project kickoff.

– Align ML targets with business questions early to avoid misalignment.

• **6.10 Decision Log:** Formally decided to report $19.1M as gross revenue and $14.44M as confirmed delivered revenue. Formally decided to exclude supplier scoring.

• **6.11 Monitoring Strategy Explanation:** The project is monitored through a dual-cadence system. Daily stand-ups track operational blockers and cross-track dependencies. Weekly variance reports track schedule and scope adherence, ensuring deviations (such as missing data fields) are escalated to the Sponsor immediately to prevent scope creep.

7. Agile Project Management

• **7.1 Product Backlog:** Contains user stories for data cleaning, dashboard creation, feature engineering, model training, RAG development, and QA.

• **7.2 Sprint Backlog:**

– Sprint 1 (Days 1-2): Data Foundation (11 tables cleaned).

– Sprint 2 (Days 3-4): Analytics & Features (Dashboards, 38-var matrix).

– Sprint 3 (Days 5-6): ML & GenAI (4 models, RAG assistant).

– Sprint 4 (Day 7): QA & Submission (24/24 tests, final report).

• **7.3 Sprint Plan:** Four rapid sprints across 7 days. Each sprint lasted 1-2 days, focusing on delivering working increments immediately to handle the strict deadline.

• **7.4 Sprint Review Template:** Each sprint ended with a demo of completed stories (e.g., showing the working dashboard or the ML model predictions) and checking against acceptance criteria.

• **7.5 Sprint Retrospective Template:** Discussed what went well, what blocked the team (e.g., missing data links), and what to change in the next sprint.

• **7.6 User Stories & Acceptance Criteria:**

– *User Story:* As a manager, I want a dashboard of revenue and delays so I can act on delivery performance. *(Criteria: Dashboard shows $19.1M gross revenue and 63.7% delay rate).*

– *User Story:* As an ML lead, I want a feature matrix so I can train models. *(Criteria: 38-variable matrix delivered with no missing values).*

• **7.7 Kanban/Scrum Board:** Tracked via Trello/Jira.



Figure 8: Kanban board (To Do, In Progress, Done)

• **7.8 Justification for Agile:** AI and analytics projects discover their true constraints as they go. Here, the missing supplier link, the missing capacity field, the revenue conflict, and the weak delay model only emerged during delivery. Short iterations with frequent reviews allowed the team to adjust scope and targets quickly, which a rigid fixed upfront plan would not have allowed.

8. Project Governance & Quality Assurance

• **8.1 Governance Structure:** Sponsor → PM → Track Leads.

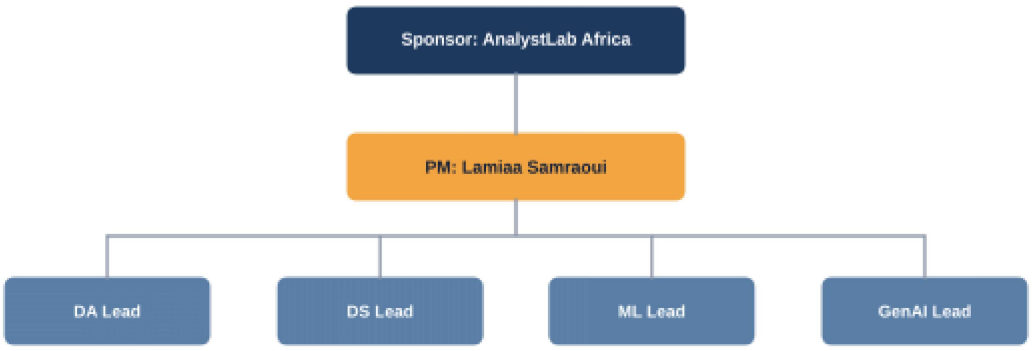


Figure 9: Governance organisation chart

• **8.2 Decision-Making Framework:** The PM has autonomy over day-to-day execution. The Sponsor must approve any scope changes or exclusions.

• **8.3 Quality Assurance Plan:** Peer review of code, Data Dictionary validation, and automated testing.

• **8.4 Quality Control Checklist:** Data validated against the dictionary. Known data issues (like cancelled shipments) disclosed on dashboards. 24/24 tests passing. Metrics reconciled or labelled.

• **8.5 Document Approval Workflow:** Track Lead creates draft → PM reviews → Sponsor approves.

• **8.6 Escalation Process:** Track Lead escalates blockers to the PM; the PM escalates unresolvable issues to the Sponsor.

• **8.7 Meeting Schedule:** Daily Slack check-ins, Weekly formal Sponsor updates.

•**8.8 Project Closure Checklist:** Deliverables complete, registers updated, repositories published, report and presentation submitted.

• **8.9 Governance Assurance Explanation:** Governance ensures successful delivery by enforcing strict data validation (the Data Dictionary) and code testing (24/24 Pytests). It prevents inaccurate reporting by forcing formal approval for scope exclusions (like supplier analysis) when the data cannot support the claims.

9. Recommendations

1. Close the Ibadan Gap: Commission a feasibility study for a local logistics hub to serve the 26.2% of customer demand currently unserved, significantly reducing the 63.7% delay rate.

2. Fix Data Foundations: Add a SupplierID field to the Products table and a WarehouseCapacity field to the Warehouse table to enable future supplier scoring and capacity analyses.

3. Standardize KPIs: Publish a single revenue definition separating gross ($19.1M$) $from\ confirmed$ (14.44M), and implement data validation rules to prevent shipments from attaching to cancelled orders.

4. Realign ML Targets: Rework delay prediction (AUC 0.52) with better operational timestamps and a clearer business target before relying on it operationally.

5. Deploy GenAI with Oversight: Roll out the RAG assistant and automated workflows to operations, while monitoring data quality at the source to prevent hallucinations.

6. Optimize Warehouse Efficiency: Utilize the DA dashboard's warehouse comparison to rebalance stock across locations (specifically Abuja W12 and Lagos W11), reducing cross-region shipment delays.

7. Improve Supplier Data Capture: Mandate the collection and linking of SupplierID on all purchase orders and shipments to enable future supplier performance scoring.

8. Reduce Product Returns: Analyze the returns dashboard to identify top-returned products and address the root causes in quality control or logistics handling.

9. Enhance Customer Retention: Use the customer analytics dashboard to identify high-value customers (like Customer 203) and design targeted loyalty interventions.

10. Establish a Performance Monitoring Cadence: Adopt the PM and DA dashboards as a monthly executive review tool to continuously track KPIs (delay rate, revenue, return rate) post-delivery.

10. Conclusion

All five tracks successfully delivered an integrated analytics, modeling, and GenAI solution that passed QA with flying colors. The project's most valuable findings are the Ibadan Deficit, the high delay rate, and the data integrity gaps. Acting on the hub study, the KPI standardization, and the data governance fixes will turn this delivery into lasting operational value for SwiftLog.